STA4001 Stochastic Processes Fall 2020
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Final Exam
Dec 25

For each question, please provide details or middle steps as much as possible.

1. (20 points) Consider a discrete time Markov Chain with state space $\{a, b, c, d, e\}$ and transition matrix

$$P = \begin{bmatrix} 0 & 0 & 0.8 & 0.2 & 0 \\ 0 & 0.4 & 0.6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0.8 & 0 & 0 & 0.2 \end{bmatrix}.$$

   (a) (2 points) What is $\mathbb{P}\{X_{2022} = b | X_{2020} = e\}$?
   (b) (3 points) What is $\mathbb{P}\{X_2 = c, X_4 = e | X_1 = b\}$?
   (c) (3 points) What is the period for state c?
   (d) (2 points) Find approximately $\mathbb{P}\{X_{100} = b | X_0 = c\}$.
   (e) (2 points) Find approximately $\mathbb{P}\{X_{100} = e | X_0 = a\}$.
   (f) (2 points) Find approximately $\mathbb{P}\{X_{100} = d | X_0 = b\}$.
   (g) (3 points) Compute $\mathbb{E}(T_b)$, the expected time for the Markov chain to first return to state b.
   (h) (3 points) Compute $\mathbb{E}(T_a)$, the expected time for the Markov chain to first return to state a.

   Solution:

   (a) $\mathbb{P}\{X_{2022} = b | X_{2020} = e\} = 0.2 \times 0.8 + 0.8 \times 0.4 = 0.48$.
   (b) $\mathbb{P}\{X_2 = c, X_4 = e | X_1 = b\} = \mathbb{P}\{X_4 = e | X_2 = c\}\mathbb{P}\{X_2 = c | X_1 = b\} = 1 \times 0.2 \times 0.6 = 0.12$.
   (c) There are 3 classes: $\{b,c,e\}$, $\{a\}$, $\{d\}$. period for state c is 1.
   (d) Easily find that $\{b,c,e\}$ are in one class; hence,their transition matrix:

$$\hat{P} = \begin{bmatrix} 0.4 & 0.6 & 0 \\ 0 & 0 & 1 \\ 0.8 & 0 & 0.2 \end{bmatrix}.$$

   Suppose the stationary distribution of $\{b,c,e\}$ is $\pi = [\pi_b, \pi_c, \pi_e]$

$$\pi\hat{P} = \pi$$
$$\text{solution:} \quad [\pi_b, \pi_c, \pi_e] = [\frac{20}{47}, \frac{12}{47}, \frac{15}{47}]$$
$$\mathbb{P}\{X_{100} = b | X_0 = c\} \approx \pi_b = \frac{20}{47}$$

   (e) $\mathbb{P}\{X_{100} = e | X_0 = a\} = \mathbb{P}\{X_{100} = e | X_1 = c\}\mathbb{P}\{X_1 = c | X_0 = a\} = \pi_e \times 0.8 = \frac{12}{47}$.
   (f) $\mathbb{P}\{X_{100} = d | X_0 = b\} = 0$.
   (g) $\mathbb{E}(T_b) = \frac{1}{\pi_b} = \frac{47}{20}$.
   (h) $\mathbb{E}(T_a) = \infty$, hence we say that it can never go back to state a.

2. (25 points) A call center is staffed by two agents, Mary and John. Mary's processing times are iid, exponentially distributed with mean 2 minutes and John's processing times are iid, exponentially distributed with mean 4 minutes.

(a) (5 points) Suppose that when you arrive, both Mary and John are busy, and you are the only one waiting in the waiting line. What is the probability that John will serve you? What is your expected time in the system (waiting plus service)?

(b) (4 points) What is the probability that you will be answered in 2 minutes?

(c) (6 points) Suppose that when you arrive, you are the 2nd one in the waiting line. What is the probability that you will be answered in 2 minutes?

(d) (10 points) Suppose the arrival process is Poisson process with rate $\lambda = 1/4$ per minute, and a call is always served by Mary if the system is empty upon arrival. Moreover, there are only three lines so that the number of calls in the system cannot exceed three. Find the long-run-average time in the system of each customer that enters the system.

Solution.

(a) From the background, we know that $T_M \sim \exp(\frac{1}{2})$, $T_J \sim \exp(\frac{1}{4})$, and T_M is independent of T_J. Hence

$$\mathbb{P}(T_M > T_J) = \frac{1/4}{1/4 + 1/2} = \frac{1}{3}.$$

And we denote the waiting time by T_{wait}, then we have that

$$T_{wait} = \min\{T_M,\ T_J\} \sim \exp(1/4 + 1/2) = \exp(3/4),\ \text{and } \mathbb{E}(T_{wait}) = \frac{4}{3},$$

hence,

$$\mathbb{E}(T_{service}) = \mathbb{E}(T_{service}|T_J < T_M)\mathbb{P}(T_J < T_M) + \mathbb{E}(T_{service}) = \mathbb{E}(T_{service}|T_J \geq T_M)\mathbb{P}(T_J \geq T_M) = \frac{8}{3}.$$

Therefore, we can conclude that $\mathbb{E}(T) = 4$.

(b) $\mathbb{P}(T_{wait} \leq 2) = 1 - e^{-3/2}$.

(c) We can view the service processes of Mary and John are two independent Poisson Processes, and denote by $\{N_M(t); t \geq 0\}$ and $\{N_J(t); t \geq 0\}$. Then we define a new Poisson Process as the summation of these two Poisson Processes, $N(t) = N_M(t) + N_J(t)$ with rate $\frac{3}{4}$. Hence, we have

$$\mathbb{P}(N(2) - N(0) \geq 2) = 1 - \frac{5}{2}e^{-3/2},$$

where $N(2) - N(0) \sim \text{Poisson}(3/2)$.

(d) We can write donw the Q matrix as follows:

$$\begin{bmatrix} -1/4 & 1/4 & 0 & 0 & 0 \\ 1/2 & -3/4 & 0 & 1/4 & 0 \\ 1/4 & 0 & -1/2 & 1/4 & 0 \\ 0 & 1/4 & 1/2 & -1 & 1/4 \\ 0 & 0 & 0 & 3/4 & -3/4 \end{bmatrix}.$$

And by solving the system

$$\begin{cases} \alpha Q = 0; \\ \sum_0^4 \alpha_i = 1, \end{cases}$$

we can obatin the long-run fraction, $\alpha = (15/28, 6/28, 3/28, 3/28, 1/28)^T$. Then, we have

$$L = \pi_M + \pi_J + 2\pi_2 + 3\pi_3 = 9/14,\ \text{and } \lambda_{eff} = \lambda(1 - \pi_3) = \frac{27}{112}.$$

Finally, according to the Little's law, we have

$$W = \frac{L}{\lambda_{eff}} = 8/3.$$

3. (15 points) Customers arrive at a single-server queue in a Poisson stream of rate λ. Each customer has a service requirement distributed as the sum of two independent exponential random variables of parameter μ. Service requirements are independent of one another and of the arrival process.

   (a) (10 points) Model the system as a CTMC by writing down the states of the system and transition rate diagram (or the generator matrix Q).

   (b) (5 points) Prove that the system is positive recurrent if and only if $\lambda/\mu < 1/2$. (You may want to consider leaving this problem to the last.)

   Solution.

   (a) We can describe the state of the system by means of the total number of remaining phases of work in the system. Each customer brings 2 phases of work to the system. State k represents there are $\lfloor \frac{k+1}{2} \rfloor$ customers in the system and $k - 2\lfloor \frac{k-1}{2} \rfloor$ remaining number of phases of the customer in service (if there is one). (Alternatively, you may describe the state of this system by means of the number of customers in the system and the remaining number of phases of the customer in service).

   The transition rates are $\lambda_{i,i-1} = \mu$ for $i \geq 1$, $\lambda_{i,i+2} = \lambda$ for $i \geq 0$. (You should draw rate diagram or write Q matrix here).

   (b) First of all, it is easy to verify that it is an irreducible DTMC. Now it suffices to prove that a stationary distribution exists if and only if $\lambda/\mu < 1/2$. Using rate in= rate out, we can derive

$$\lambda\pi_0 = \mu\pi_1, \ (\lambda + \mu)\pi_n = \lambda\pi_{n-2} + \mu\pi_{n+1}$$

   for $n \geq 1$ (let $\pi_{-1} = 0$). Using guess & check method, we propose $\pi_n = x^n$ and plug into the above equation, it becomes $(\lambda + \mu)x^2 = \lambda + \mu x^3$. There is a trivial solution $x = 1$, by excluding that we obtain $\mu x^2 - \lambda x - \lambda = 0$. It is easy to see that there is a root in $(0, 1)$ to this equation is equivalent to $\lambda/\mu < 1/2$, completing the proof. (For more rigorous derivations, one should apply the method of moment generating function, which is not required here).

4. (20 points) Let $S_n = X_1 + \cdots + X_n$ be a symmetric simple random walk starting at 0 and let $T = \inf\{n : S_n \notin (-a, a)\}$ where a is a positive integer.

   (a) (6 points) Prove that T is a stopping time with respect to $X_1, \cdots, X_n$.

   (b) (7 points) Use the fact that $S_n^2 - n$ is a martingale to show that $\mathbb{E}T = a^2$.

   (c) (7 points) Find constant b so that $Y_n = S_n^4 - 6nS_n^2 + 3n^2 + bn$ is a martingale, and use this to compute $\mathbb{E}T^2$.

   For (b) and (c), please justify your derivation in every step by quoting the appropriate results presented in lectures.

   Solution.

   (a) By defintion of stopping time, we need to show that for any $n, \{T = n\}$ (or $\{T \leq n\}$) can be determined by $X_1, \ldots, X_n$ Note that $S_1, \ldots, S_n$ can be determined by $X_1, \ldots, X_n$, so $\{T = n\} = \{S_1 \in (-a, a), \cdots, S_{n-1} \in (-a, a), S_n \notin (-a, a)\}$ can be determined by $X_1, \ldots, X_n$, i.e., given the value of $X_1, \ldots, X_n$ we can tell whether $T = n$ or not. It follows that T is a stopping time. Notes: two common mistakes: (1) show that T can be determined by $X_1, \ldots, X_n$, this is not correct defintion; (2) show that T is the first hitting time of $\{-a, a\}$ therefore is a stopping time. since this question requires proof, you can only quote appropriate theoretical results (theorems, lemmas, etc), not examples in the lecture notes (first passage time as a stopping time is an example in lecture notes).

(b) Apply Optional Theorem to $S_n^2 - n$ with stopping time $T \wedge n$, we have

$$\mathbb{E}\left(S_{T\wedge n}^2 - T \wedge n\right) = 0$$

Therefore,

$$\mathbb{E}\left(S_{T\wedge n}^2\right) = \mathbb{E}(T \wedge n)$$

Take limit as $n \to \infty$ for both sides,

$$\lim_{n\to\infty} \mathbb{E}\left(S_{T\wedge n}^2\right) = \lim_{n\to\infty} \mathbb{E}(T \wedge n)$$

For l.h.s, notice that,

$$P\left(\lim_{n\to\infty} S_{T\wedge n}^2 = S_T^2\right) = 1, \text{ a.s.}$$

and

$$\left|S_{T\wedge n}^2\right| \leq a^2$$

Thus we could apply dominated convergence theorem, and have

$$\lim_{n\to\infty} \mathbb{E}\left(S_{T\wedge n}^2\right) = \mathbb{E}\left(\lim_{n\to\infty} S_{T\wedge n}^2\right) = \mathbb{E}\left(S_T^2\right)$$

For r.h.s, notice that,

$$P\left(\lim_{n\to\infty} T \wedge n = T\right) = 1, \text{ a.s.}$$

and

$$T \wedge 0 \leq T \wedge 1 \leq T \wedge 2 \leq \cdots, \text{ for } n = 0, 1, \cdots$$

Thus we could apply monotone convergence theorem, and have

$$\lim_{n\to\infty} \mathbb{E}(T \wedge n) = \mathbb{E}\left(\lim_{n\to\infty} T \wedge n\right) = \mathbb{E}(T)$$

Hence we conclude that

$$\mathbb{E}(T) = \mathbb{E}\left(S_T^2\right) = a^2$$

(c) First we derive constant b.

$$\mathbb{E}\left[Y_{n+1} \mid X_1, \cdots, X_n\right] = Y_n$$

when $b = 2$. Therefore $\{Y_n\}$ is a stopping time when $b = 2$. Apply Optional Theorem to Y_n with stopping time $T \wedge n$,

$$\mathbb{E}S_{n\wedge T}^4 + b\mathbb{E}(n \wedge T)^2 + 2\mathbb{E}n \wedge T = 6\mathbb{E}\left(n \wedge TS_{n\wedge T}^2\right)$$

Take limit as $n \to \infty$ for both sides,

$$\lim_{n\to\infty} \left[\mathbb{E}S_{n\wedge T}^4 + 3\mathbb{E}(n \wedge T)^2 + 2\mathbb{E}n \wedge T\right] = 6 \lim_{n\to\infty} \left[\mathbb{E}\left(n \wedge TS_{n\wedge T}^2\right)\right]$$

For l.h.s, notice that,

$$P\left(\lim_{n\to\infty} S_{T\wedge n}^2 = S_T^2\right) = 1, \text{ a.s.}$$

and

$$\left|S_{T\wedge n}^4\right| \leq a^4$$

Thus we could apply dominated convergence theorem, and have

$$\lim_{n\to\infty} \mathbb{E}\left(S_{T\wedge n}^4\right) = \mathbb{E}\left(\lim_{n\to\infty} S_{T\wedge n}^4\right) = \mathbb{E}\left(S_T^4\right) = a^4$$

Also notice that

$$P\left(\lim_{n\to\infty} T \wedge n = T\right) = 1, \text{ a.s.}$$

and

$$T \wedge 0 \leq T \wedge 1 \leq T \wedge 2 \leq \cdots, \text{ for } n = 0, 1, \cdots$$

Thus we could apply monotone convergence theorem, and have

$$\lim_{n\to\infty} \mathbb{E}(T \wedge n) = \mathbb{E}(T), \lim_{n\to\infty} \mathbb{E}(T \wedge n)^2 = \mathbb{E}\left(T^2\right)$$

For r.h.s., note that $n \wedge TS^2_{n\wedge T} \leq a^2 T$ for any n and $\mathbb{E}T < \infty$ by part (b), we could apply dominated convergence theorem and

$$6 \lim_{n\to\infty} \left[\mathbb{E}\left(n \wedge TS^2_{n\wedge T}\right)\right] = 6\mathbb{E}\left(TS^2_T\right) = 6a^2\mathbb{E}T = 6a^4$$

In conclusion we have $\mathbb{E}T^2 = \left(5a^4 - 2a^2\right)/3$.

5. (20 points) Consider a special two-station open Jackson network. Each station has a single server and an infinite capacity waiting area. External arrivals to station 1 follow a Poisson process with rate α, and service times at station i are iid, following exponential distribution with rate μ_i, $i = 1, 2$. After being served at station 1, each customer goes to station 2, and after being served at station 2, the customer goes back to station 1 with probability 20% and leaves the system with probability 80%.

   Assume that

   $$(1.25)\alpha < \mu_1 \text{ and } (1.25)\alpha < \mu_2.$$

   Let $X_i(t)$ be the number of customers at station i at time t, $i = 1, 2$. Then $X = \{(X_1(t), X_2(t)), t \geq 0\}$ is a continuous time Markov chain.

   Show that X is positive recurrent by proving that

   (a) (5 points) X is irreducible,

   (b) (15 points) X has a stationary distribution.

   For (b), do the following three steps: (i) write the flow balance equation for *each state*, (ii) guess a form of the stationary distribution, and (iii) verify that the guessed stationary distribution does satisfy the flow balance equation for each state.

Solution:

We first prove that X is irreducible. For that, we need to prove that for any state (n_1, n_2), it can reach any other state (m_1, m_2) with a positive probability. We prove this statement by proving that (n_1, n_2) can reach state $(0, 0)$ and that state $(0, 0)$ can reach state (m_1, m_2). We first prove that state (n_1, n_2) can reach state $(0, 0)$. We illustrate this proof by considering state $(n_1, n_2) = (2, 3)$. The following diagram proves that state $(2, 3)$ can reach state $(0, 0)$ in positive probability:

$$(2,3) \overset{\mu_1}{\to} (1,4) \overset{\mu_1}{\to} (0,5) \overset{.8\mu_2}{\to} (0,4) \overset{.8\mu_2}{\to} (0,3) \overset{.8\mu_2}{\to} (0,2) \overset{.8\mu_2}{\to} (0,1) \overset{.8\mu_2}{\to} (0,0)$$

The following diagram illustrates the proof that state $(0, 0)$ can reach state (m_1, m_2) by choosing $(m_1, m_2) = (2, 1)$:

$$(0,0) \overset{\alpha}{\to} (1,0) \overset{\alpha}{\to} (2,0) \overset{\alpha}{\to} (3,0) \overset{\mu_1}{\to} (2,1).$$

Let $\pi = \{\pi(n_1, n_2) : n_1, n_2 \in \mathbb{Z}_+\}$ be the stationary distribution.

(a) For each state (n_1, n_2), π must satisfies the following balance equation: the flow rate out of the state is equal to the flow rate into the state. We consider for sets of states.

   (i) $(n_1, n_2) = (0, 0)$: The balance equation is

   $$\alpha\pi(0, 0) = .8\mu_2\pi(0, 1). \tag{1}$$

(ii) $(n_1, n_2) = (n_1, 0)$ with $n_1 > 0$: The balance equation is

$$(\alpha + \mu_1)\pi(n_1, 0) = \alpha\pi(n_1 - 1, 0) + .8\mu_2\pi(n_1, 1) + .2\mu_2\pi(n_1 - 1, 1). \tag{2}$$

(iii) $(n_1, n_2) = (0, n_2)$ with $n_2 > 0$: The balance equation is

$$(\alpha + \mu_2)\pi(0, n_2) = \mu_1\pi(1, n_2 - 1) + .8\mu_2\pi(0, n_2 + 1). \tag{3}$$

(iv) $(n_1, n_2) = (n_1, n_2)$ with $n_1 > 0$ and $n_2 > 0$: The balance equation is

$$\begin{aligned}(\alpha + \mu_1 + \mu_2)\pi(n_1, n_2) &= \alpha\pi(n_1 - 1, n_2) + \mu_1\pi(n_1 + 1, n_2 - 1) \\ &\quad + .8\mu_2\pi(n_1, n_2 + 1) + .2\mu_2\pi(n_1 - 1, n_2 + 1).\end{aligned} \tag{4}$$

(b) We conjecture that

$$\pi(n_1, n_2) = (1 - \rho_1)\rho_1^{n_1}(-\rho_2)\rho_2^{n_2} \quad \text{for } n_1, n_2 \in \mathbb{Z}_+, \tag{5}$$

where $\rho_i = \lambda_i/\mu_i$, $i = 1, 2$, and (λ_1, λ_2) satisfies

$$\begin{aligned}\lambda_1 &= \alpha + .2\lambda_2, \\ \lambda_2 &= \lambda_1.\end{aligned}$$

Solving the linear system of equations we have $\lambda_1 = \lambda_2 = 1.25\alpha$.

(c) We now verify that π, defined in (5), satisfies (1)-(4). We first verify (4). Equation (4) is equivalent to

$$\begin{aligned}(\alpha + \mu_1 + \mu_2)(1 - \rho_1)\rho_1^{n_1}(1 - \rho_2)\rho_2^{n_2} &= \alpha(1 - \rho_1)\rho_1^{n_1 - 1}(1 - \rho_2)\rho_2^{n_2} \\ &\quad + \mu_1(1 - \rho_1)\rho_1^{n_1 + 1}(1 - \rho_2)\rho_2^{n_2 - 1} + .8\mu_2(1 - \rho_1)\rho_1^{n_1}(1 - \rho_2)\rho_2^{n_2 + 1} \\ &\quad + .2\mu_2(1 - \rho_1)\rho_1^{n_1 - 1}(1 - \rho_2)\rho_2^{n_2 + 1},\end{aligned}$$

which is further equivalent to

$$(\alpha + \mu_1 + \mu_2)\rho_1\rho_2 = \alpha\rho_2 + \mu_1\rho_1^2 + .8\mu_2\rho_1\rho_2^2 + .2\mu_2\rho_2^2.$$

Since $\lambda_1 = \lambda_2$, $\rho_i = \lambda_1/\mu_i$ for $i = 1, 2$. Therefore, the preceding equation is equivalent to

$$(\alpha + \mu_1 + \mu_2)(\lambda_1/\mu_1)(\lambda_1/\mu_2) = \alpha(\lambda_1/\mu_2) + \mu_1(\lambda_1/\mu_1)^2 + .8\mu_2(\lambda_1/\mu_1)(\lambda_1/\mu_2)^2 + .2\mu_2(\lambda_1/\mu_2)^2,$$

or

$$(\alpha + \mu_1 + \mu_2)\lambda_1 = \alpha\mu_1 + \lambda_1\mu_2 + .8\lambda_1^2 + .2\lambda_1\mu_1,$$

or

$$(\alpha + \mu_1)\lambda_1 = \alpha\mu_1 + .8\lambda_1^2 + .2\lambda_1\mu_1.$$

Since $\lambda_1 = \alpha + .2\lambda_1$, we have $\lambda_1\mu_1 = \alpha\mu_1 + .2\lambda_1\mu_1$. Thus the preceding equation is reduced to

$$\alpha\lambda_1 = .8\lambda_1^2,$$

which holds because $\alpha = .8\lambda_1$.

Equations (1)-(3) can be verified similarly.